

Subject Description Form

Subject Code	EIE4124
Subject Title	Modern Robotics
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: EIE3312 Linear Systems (A waiver will be given to senior year intake students, subject to their academic backgrounds.)
Objectives	Robots have been in our daily lives, integrating seamlessly into many fields. They play significant roles in the new technological revolution. This subject objective is to introduce in a systematic manner the advanced technologies used for modern robotic applications.
Intended Subject Learning Outcomes	Upon completion of the subject, students will be able to: <u>Category A: Professional/academic knowledge and skills</u> 1. Appreciate hardware and software of modern robotics. 2. Appreciate the different techniques for autonomous robots. 3. Understand some technologies for embodied intelligence in robotics. <u>Category B: Attributes for all-roundedness</u> 4. Communicate effectively with others on the acquired knowledge. 5. Appreciate the importance of staying abreast of the state-of-the-art technologies.
Subject Synopsis/ Indicative Syllabus	1. <u>Introduction of Robotic Platforms and Relevant Hardware</u> • Examples of humanoid robots, industrial robots for manufacturing and robotic platforms for research. • Introduction of robotic hardware: sensors, motors, actuators, end effectors, and connectors. 2. <u>Software architecture for modern robotics</u> • Introduction of robotic software: robot operating system, digital twin, robot learning and thinking • Demonstrations from Dassault Systèmes and NVIDIA. 3. <u>Intelligent Control and Autonomous Robots</u> • Methods for intelligent control of robots • Studies of autonomous robot systems 4. <u>Embodied Intelligence in Modern Robotics</u> • Technologies of embodied intelligence in modern robotics • Prospective of embodied large language models enabled robots. 5. <u>Object Detection and Tracking</u> • Object recognition and tracking with visual sensors for moving object. 6. <u>Visual Tracking Control</u> • Camera-based tracking control 7. <u>Body Motion Capture Technologies</u> • Body motion tracking with a Kinect sensor.

	8. <u>Obstacle Avoidance for Robot Manipulator</u> Obstacle avoidance strategy and collision prediction algorithm.																																																				
Teaching/Learning Methodology	Lectures and Tutorials: The subject matters will be delivered through lectures. Students will be engaged in the tutorials through Q&A, discussions, and other activities. Laboratory Activities: Students will do experiments using software tools. Students will appreciate the technologies in modern robotics that they have learned from lectures and put them into practice in simulation environments. Test: Students will finish class exercises by doing self-studies.																																																				
Assessment Methods in Alignment with Intended Subject Learning Outcomes	<table border="1"> <thead> <tr> <th rowspan="2">Specific Assessment Methods/Tasks</th><th rowspan="2">% Weighting</th><th colspan="5">Intended Subject Learning Outcomes to be Assessed (Please tick as appropriate)</th></tr> <tr> <th>1</th><th>2</th><th>3</th><th>4</th><th>5</th></tr> </thead> <tbody> <tr> <td>1. Continuous Assessment</td><td></td><td></td><td></td><td></td><td></td><td></td></tr> <tr> <td>• Laboratory activities</td><td>20%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr> <tr> <td>• Tests</td><td>30%</td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td></tr> <tr> <td>2. Examination</td><td>50%</td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td></tr> <tr> <td>Total</td><td>100%</td><td colspan="5"></td></tr> </tbody> </table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Laboratory activities will require students to apply what they have learned to solve problems. There will be questions that allow students to exercise their creativity in design and implementation through discussions and practicing, and appreciate the need for knowing the robotic technologies. Tests and Examination assess students' achievement of the learning outcomes in a more formal manner. 100% of the assessment is individual assessment.						Specific Assessment Methods/Tasks	% Weighting	Intended Subject Learning Outcomes to be Assessed (Please tick as appropriate)					1	2	3	4	5	1. Continuous Assessment							• Laboratory activities	20%	✓	✓	✓	✓	✓	• Tests	30%	✓	✓	✓			2. Examination	50%	✓	✓	✓			Total	100%					
Specific Assessment Methods/Tasks	% Weighting	Intended Subject Learning Outcomes to be Assessed (Please tick as appropriate)																																																			
		1	2	3	4	5																																															
1. Continuous Assessment																																																					
• Laboratory activities	20%	✓	✓	✓	✓	✓																																															
• Tests	30%	✓	✓	✓																																																	
2. Examination	50%	✓	✓	✓																																																	
Total	100%																																																				
Student Study Effort Expected	Class contact (time-tabled):																																																				
	• Lectures					22 Hours																																															
	• Tutorial/Laboratory					17 Hours																																															
	Other student study effort:																																																				
	• Lecture: preview/review of notes; preparation for test					36 Hours																																															
	• Tutorial/Laboratory: preview of materials, revision and/or reports writing					30 Hours																																															
	Total student study effort:					105 Hours																																															
Reading List and References	Reference Materials: C. Yang, H. Ma, and M. Fu, <i>Advanced Technologies in Modern Robotic Applications</i>. Singapore: Springer Singapore Pte. Limited, 2016. R. Murphy, <i>Introduction to AI Robotics (2nd Ed.)</i>. Cambridge, Massachusetts: The MIT Press, 2019. A. Koubaa (editor), <i>Robot Operating System (ROS): The Complete Reference (Volume 5)</i>. Cham: Springer International Publishing: Imprint: Springer, 2021.																																																				

	4. Fouad Sabry, Autonomous Robot, Advancements in Intelligent Systems for Self Guided Machines, One Billion Knowledgeable, 2025 5. Huaping Liu, Xinzhu Liu, Kangyao Huang, Di Guo, Embodied Multi-Agent Systems: Perception, Action, and Learning, Springer Nature, 2025
--	--

July 2025